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Attentional Transplants for Industrial Applications

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ABSTRACT

Here we propose a new design strategy for mixed and virtual reality designers, which we call 'attentional transplants'. The strategy is to direct a user's visual attention in a way which mirrors the attention of a previous user, in order to produce similar affective or cognitive responses in the second user. Attentional transplants may be useful across a wide range of design situations in which one's manner of attending reliably drives affective and cognitive outcomes. In order to test whether transplanting one participant's pattern of attention into another can produce similar downstream responses in the second participant, we ran two online experiments. In Experiment 1a, 50 observers viewed images of landscapes by using their cursor to move a small circular viewing window around each image for three seconds. After viewing an image, they rated how much they liked it. For each image, we identified two 'attentional donors' — the Liked-it-Most observer who rated the image highest (standardized relative to their other image ratings) across observers, as well as the Liked-it-Least observer who rated the image lowest across observers. In Experiment 1b, we recruited 100 new observers to serve as 'attentional recipients'. These observers viewed each image, but now passively, through a moving window which reproduced the viewing pattern of either the Experiment 1 observer who Liked it Most, or the observer who Liked it Least. Recipients gave substantially higher ratings to an image when they received the viewing pattern of the observer who Liked it Most, compared to when they received the viewing pattern of the observer who Liked it Least. Based on this success, we speculate about several additional ways in which attentional transplants could be used in virtual and mixed reality systems, in order to achieve desired cognitive and affective outcomes.

Keywords: Visual attention, Virtual reality, Extended reality.

INTRODUCTION

Visual attention, the selection of visual information for a task, is a crucial leverage point for designers. How we attend determines what we notice (e.g. Mack & Rock, 1998), which pieces of visual information appear to "go together" (e.g. Scholl, 2001), and whether we encode that information into long-term memory (e.g. Chun & Turk-Browne, 2007). Research in visual cognition has confirmed a number of general principles for guiding attention (e.g. if you want users of a digital interface to focus on a particular icon, have it suddenly start to move; Abrams & Christ, 2003). However, new drivers of visual attention are continually being discovered (e.g. Nguyen & van Buren, 2024), and it is unclear whether vision science will uncover an exhaustive list of principles.

Here we propose a powerful new approach to designing for attention, which XR designers can use to orient users' attention in a theory-free, *outcome-driven* way. Step 1 is to record the pattern of visual attention from a 'donor' who shows a desirable outcome (e.g. enjoying using a visual interface, or succeeding at using it to perform some task). Step 2 is to cue a 'recipient' to attend in the same way, in order to cause them to achieve the same outcome. This method of 'attentional transplants' is useful across a wide range of design situations in which users' manner of attending reliably drives affective and cognitive outcomes. We first describe a proof-of-principle that attentional transplants are effective at causing recipients to enjoy images similarly to donors. We then describe several new domains in which we are currently applying this method, with the help of AR and VR technology.

PROOF-OF-PRINCIPLE: ATTENTIONAL TRANSPLANTS CAUSE RECIPIENTS TO LIKE IMAGES SIMILARLY TO DONORS

We first describe two online experiments, which we ran in order to test whether transplanting one participant's pattern of attention into another can produce similar downstream responses in the second participant — here focusing on *preferences for what one is seeing*. In both experiments, we asked one group of 'donor' participants to move a small circular window to view images. Such mouse-based visual information selection correlates well with eye-tracking measures of attention (Egner et al., 2018), and meets the most commonly used definition of visual attention, as visual information selection for a task (Wu, 2014; 2023). After viewing each image, the participant rated how much they liked it. For each image, we identified the participant who liked the image most (i.e. who provided the highest standardized preference rating for the image, relative to all participants in the sample), and the participant who liked the image least. We then showed a new group of 'recipient' observers the same images, through a moving viewing window which replicated the viewing pattern of either the Liked-it-Most participant, or the Liked-it-Least participant. We reasoned that, if attention drives preferences in a similar way across observers, recipients would tend to prefer images viewed with the attentional patterns of Liked-it-Most participants, relative to images viewed with the attentional patterns of Liked-it-Least participants. All design and analysis procedures were preregistered at https://osf.io/akymp/.

Experiment 1a: Identifying attentional donors

In Experiment 1a, 50 participants viewed 50 landscape images by using their cursor to move a small circular viewing window around each image for three seconds. After viewing each image, they used

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a slider to rate how much they liked it from -100 (strongly dislike) to 100 (strongly like). For each image, we identified two 'attentional donors' — the Liked-it-Most participant who rated the image highest (standardized relative to their other image ratings) across participants, as well as the Liked-it-Least participant who rated the image lowest across participants. Figure 1a depicts, for a single image, the scanpath of the Liked-it-Most participant (in green), and the scanpath of the Liked-it-Least participant (in red).

Experiment 1b: Do Attentional Transplants Influence Recipients' Preferences?

In Experiment 1b, we recruited 100 new observers to serve as 'attentional recipients'. These observers viewed each image, but in this experiment, instead of moving the viewing aperture themselves, observers viewed a passive "replay" of an Experiment 1 participant's viewing trajectory for the image. Half of the trials were Liked-it-Most trials, in which the observer saw the image via the viewing trajectory of the Experiment 1 participant who liked the image the most, and the other half of trials were Liked-it-Least trials, in which the observer saw the image via the viewing trajectory of the Experiment 1 participant who liked the image the least. To counterbalance the assignment of the 50 images to the Liked-it-Most and Liked-it-Least conditions, the images were randomly split into Image Group A (25 images) and Image Group B (25 images). Half of the recipient observers viewed Image Group A with Liked-it-Most viewing paths, and Image Group B with Liked-it-Least viewing paths; the other half of recipient observers viewed Image Group A with Liked-it-Least viewing paths, and Image Group B with Liked-it-Most viewing paths.

After viewing each image, recipient observers rated the image from -100 (strongly dislike) to 100 (strongly like). We hypothesized that recipient observers would give significantly higher standardized ratings to images that they viewed via Liked-it-Most trajectories, compared to images that they viewed with Liked-it-Least trajectories.

Results

Figure 1b depicts the average ratings that recipients gave to images viewed via Liked-it-Most vs. Liked-it-Least viewing trajectories. Inspection of the figure suggests that recipients gave substantially higher ratings to images which they viewed via the viewing pattern of the donor who Liked it Most (M=58.04), compared to images which they viewed via the viewing pattern of the donor who Liked it Least (M=-41.69), $t(99)=8.49$, $p<.001$, $d_z=0.69$.

Direct Replication

Given the importance of direct replications, we next reran the study with a new sample of 50 attentional donors, and a new sample of 100 attentional recipients. Figure 1c depicts the average ratings that recipients gave to images viewed via Liked-it-Most vs. Liked-it-Least viewing trajectories. Inspection of the figure suggests that recipients again gave higher ratings to images which they viewed via the viewing pattern of the donor who Liked it Most (M=65.75), compared to images which they viewed via the viewing pattern of the donor who Liked it Least (M=-52.25), $t(99)=11.22$, $p<.001$, $d_z=0.79$.

These experiments provide an initial proof-of-principle that the manner in which one attends to an image drives how much one likes it, that there are inter-observer similarities in how different patterns of attention drive preferences, and that designers may thus use attentional transplants in order to help recipient observers enjoy images more.

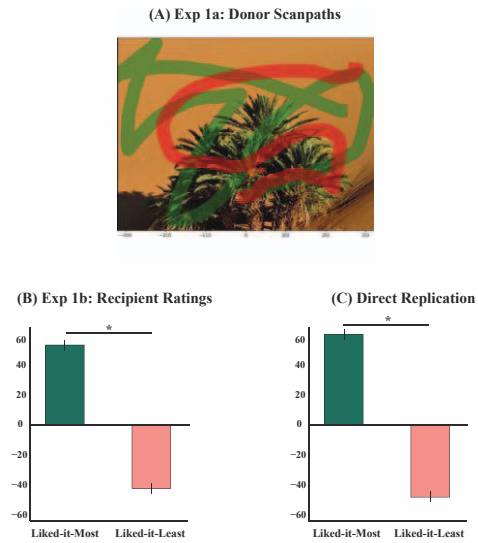

Figure 1: (A) For a single image in Experiment 1a, depiction of the scanpath of the Liked-it-Most participant (in green, preference rating = 44.82), and the scanpath of the Liked-it-Least participant (in red, preference rating = -38.10). (B) Recipient observers' average preference ratings for images viewed with Liked-it-Most and Liked-it-Least scanpaths. (C) Recipient observers' average ratings in the direct replication. Error bars reflect 95% confidence intervals, subtracting out the shared variance.

USING ATTENTIONAL TRANSPLANTS TO INFLUENCE HEDONIC RESPONSES IN VR AND MIXED REALITY ENVIRONMENTS

Based on the success of these experiments we are now designing a series of experiments to test whether attentional transplants can be used to achieve hedonic and performance outcomes in VR and mixed reality environments. For example, in close analogy to the preceding experiments, in both a virtual gallery (using a VR headset with eye-tracking capability) and a real art gallery (viewed through AR glasses with eye-tracking capability), we are planning to record users' fixation patterns while they view paintings. We will run experiments which are similar to the above, in order to determine whether transplanting the fixation patterns of a 'donor' gallery visitor into a 'recipient' visitor can cause the recipient to like, or dislike, a painting.

Whereas the preceding experiments transplanted *spatial* attention by replaying the trajectory of a moving viewing window, these experiments will infer and transplant *object-based* attention. Specifically, we will use a visual segmentation algorithm to parse each painting into bounded object regions (Chen et al., 2018). For both Liked-it-Most donors and Liked-it-Least donors, we will record the proportion of time that the donor fixates within each region. When recipients view the painting in VR or AR glasses, we will blur the entire painting, and unblur different regions in proportion to the donor's fixation duration. We will also explore other methods of orienting attention to these donor-attended regions, e.g. by rendering them in color and desaturating unattended regions, or (if necessary) by introducing slight motion transients to more robustly attract the recipient's attention to donor-

attended regions. With this general approach, we will be able to attract a recipient's attention to regions in the painting in a way which reflects the donor's attention, and which is robust to viewpoint differences. As in our experiments above, we predict that in these new VR and mixed reality implementations, attentional transplants from Liked-it-Most donors will cause recipients to like paintings more, and attentional transplants from Liked-it-Least donors will cause recipients to like paintings less, as assessed through preference ratings.

The beauty of the attentional transplants approach is that it can be applied to any outcome measure that one expects is influenced by attention in a similar way across different observers — and that this method can be used to achieve desirable outcomes without needing to theorize much at all about *why* particular attention patterns drive those outcomes. To demonstrate the general utility of attention transplants, we are also planning experiments to determine whether attentional transplants from a donor who found a painting highly moving make recipients find the painting more moving, and whether attentional transplants from a donor who found a painting highly interesting make recipients find the painting more interesting. Beyond paintings, we also predict that attentional transplants are effective at altering hedonic responses for a much broader range of stimuli — e.g. store window displays, scenic lookouts, building facades, advertisements, etc. (In all of these cases, we acknowledge that attention is only one factor that determines how much we enjoy looking at something — but as you can see from the results above, it is a *big* factor!)

USING ATTENTIONAL TRANSPLANTS TO IMPROVE PERFORMANCE IN VR AND MIXED REALITY ENVIRONMENTS

Of course visual attention is important not only to subjective enjoyment, but also to situational awareness, and task performance (e.g. noticing and avoiding pedestrians while driving; Kim & Gabbard, 2020). We are thus also planning a series of experiments to test whether attentional transplants can be used to improve performance in a variety of real-world tasks, including awareness of pedestrians in a driving simulation, target recognition in police training, and detection of tumors in radiological scans — including separate experiments to determine whether we can transplant attention in order to influence both recipients' sensitivity, as well as their response bias.

In many tasks, successful performance requires us to attend in new ways relative to how we have attended before. (For example, when searching for our keys, we are less likely to find them if we keep searching in the same locations.) Thus, we plan to test another class of applications, in which we will record a perceiver's own pattern of attention from the past, in order to generate a *maximally different* pattern of attention, and to transplant this into them (a kind of assisted 'inhibition of return'; Klein, 2000). Transplanting novel patterns of attention may not only help searchers to find their search targets; it may also turn out to be useful in a broader range of cognitive tasks, e.g. in order to help problem-solvers find new solutions in creative tasks (e.g. Weatherford et al., 2021).

CONCLUSION

Here we have described the overall logic of attentional transplants, and discussed several ways in which we hope that they can be useful for industrial applications. This method has both strengths (e.g. one can use it without needing a complete theory of how attention is driving the desired outcome) and limitations (e.g. it will not work when the outcome does not depend on how one attends, or when it does, but not in a way which is consistent across different individuals). As a proof-of-principle, we have shown that transplanting the pattern of attention from a donor who enjoyed an image causes a recipient to enjoy viewing the image more, and that transplanting the pattern of attention from a donor who disliked an image causes a recipient to like the image less. We are excited about the broad applicability of this technique, particularly when combined with virtual and mixed reality technology, and connected to different outcome measures, including industrial safety outcomes.

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